Timothy Belin

Mechanical Engineering Ph.D. Student | University of California, Berkeley

720-385-5319 | timbelin@berkeley.edu | LinkedIn | Google Scholar

EDUCATION

University of California, Berkeley — Berkeley, CA

Ph.D. Mechanical Engineering — *May 2031 (Expected)*

M.Sc. Mechanical Engineering — *May 2028 (Expected)*

Union College — Schenectady, NY

B.S. Mechanical Engineering (Honors) — *June 2026*

Summa Cum Laude

RESEARCH EXPERIENCE

Research Interests

Multiphase flows · air lubrication · wall-bounded turbulence · experimental flow measurement techniques

Graduate Student Researcher – Associate Professor Simo Mäkiharju — Aug. 2026 – Present

University of California, Berkeley — *Berkeley, CA*

- Experimentally investigates air-lubricated multiphase flows for frictional drag reduction, characterizing how gas injection modifies the structure of the air-water interface and underlying turbulent boundary layer.
- Maintains and operates a recirculating multiphase flow channel; collects particle image velocimetry (PIV) measurements to extract mean flow fields and turbulence statistics.
- Designing a multi-camera tomographic particle tracking velocimetry (PTV) setup for three-dimensional flow diagnostics.

Undergraduate Research Assistant – Associate Professor Ali Hamed — June 2024 - August 2026

Union College — *Schenectady, NY*

- Experimentally investigated turbulent boundary-layer flow over rough surfaces to determine the effects of the roughness geometry on the mean flow characteristics and turbulence structure
- Employed planar PIV and volumetric PTV in a recirculating water tunnel and an open-return wind tunnel for two- and three-dimensional flow field measurements.
- Developed MATLAB workflows for processing, analyzing, and visualizing mean flow fields, turbulence quantities, and second-order turbulence statistics.
- Drafted journal publication manuscripts and assisted responses to peer-reviews.

PUBLICATIONS

Peer-reviewed Manuscripts

[2] **T. R. Belin**, B. Mlavsky, and A. M. Hamed (2025). Turbulent flow past a finite wall-mounted cylinder at various degrees of upstream sheltering: Effects of the number of upstream cylinders. *Physics of Fluids*, 37(11), Art. No. 115158, 2025.

[1] A. M. Hamed, I. A. Spoor, B. R. McAtee, and **T. R. Belin** (2025). Vortical structures past isolated and sheltered wall-mounted cylinders immersed in a turbulent boundary layer. *Experiments in Fluids*, 66(12), 212, Art. No. 212, 2025.

Manuscripts in Preparation

[*] **T. R. Belin**, W. H. Good, J. A. Carreiro, M. Kaid, and A. M. Hamed Wake organization and flow relaxation past rigid and flexible isolated roughness elements.

[*] A. M. Hamed, **T. R. Belin**, B. R. McAtee, and Z. V. Panlilio Effects of spanwise offset on the flow and vortical structures in the wake of sheltered finite wall-mounted circular cylinders.

CONFERENCE PRESENTATIONS

- **T. R. Belin**, B. R. McAtee, Z. V. Panlilio, and A. M. Hamed, "Flow and vortical structures past sheltered finite wall-mounted cylinders: effects of spanwise offset," American Physical Society – Division of Fluid Dynamics, Orlando, FL (November, 2026).
- **T. R. Belin**, B. Mlavsky, and A. M. Hamed, "Impact of the number of upstream sheltering cylinders on the wake of a downstream finite wall-mounted cylinder," American Physical Society – Division of Fluid Dynamics, Houston, TX (November, 2025).
- B. Mlavsky, **T. R. Belin**, and A. M. Hamed, "Turbulent flow past a low-aspect-ratio finite wall-mounted cylinder at various degrees of upstream sheltering," American Physical Society – Division of Fluid Dynamics, Salt Lake City, UT (November, 2024).

TEACHING EXPERIENCE

Supplemental Instructor Apr. 2024 – Jun. 2025
Union College Mechanical Engineering Department *Schenectady, NY*

- Supported student learning in Programming and Writing in Mechanical Engineering, a second-year mechanical engineering course teaching library research, scientific writing, and data analysis skills.
- Led review sessions for 25 students on MATLAB coding, including visualization of large datasets, numerical differentiation/integration, and numerical solutions to simple differential equations.
- Gave feedback to students in individual appointments on the clarity, conciseness, and adherence to scientific writing conventions within their writing assignments.

Peer Writing Tutor Sep. 2023 – Mar. 2024
Union College Writing Center *Schenectady, NY*

- Provided individual guidance on essays, lab reports, and other forms of written communication.
- Assisted students with organization, argument structure, and formatting.

LEADERSHIP, SERVICE & OUTREACH

Latitude High School Sep. 2026 - Present
Volunteer Computer Science Mentor *Oakland, CA*

- Attends student project presentation sessions as a volunteer programming professional
- Provides feedback to students on the design and implementation of their programming projects

Union College Pi Tau Sigma Mechanical Engineering Honor Society Apr. 2025 - Apr. 2026
Vice President *Schenectady, NY*

- Organized the Union College mechanical engineering help desk, which included coordinating member volunteer schedules and collaborating with the mechanical engineering department faculty.
- Provided extracurricular academic support to mechanical engineering students on homework, laboratory reports, and other technical assignments.

Union College Ultimate Frisbee Club Team Jun. 2025 - May 2026
President *Schenectady, NY*

- Coordinated team practices, tournament travel, registration, and logistics. Served as liaison between the team, the Student Activities department, and Finance Committee.
- Fostered a positive team culture by supporting player development and ensuring an inclusive environment on and off the field for all team members.

Union College Outing Club Apr. 2025 - Apr. 2026
President *Schenectady, NY*

- Led hiking, rock climbing, white water rafting, and mountain biking trips for other students to enjoy and discover the beauty of local outdoor settings.
- Organized transportation, funding, and community outreach for each outdoor trip in collaboration with the Student Activities Department and the other Outing Club student leaders.

Union College Science and Technology Entry Program Oct. 2022- Jun. 2023
Volunteer Tutor *Schenectady, NY*

- Tutored math and science for underserved high school students from the local Schenectady school system.

Lafayette Public Library Homework Center Sep. 2019- Mar. 2020
Volunteer Tutor *Lafayette, CO*

- Provided support in math, science, reading, and writing to local elementary and middle school students

OTHER PROFESSIONAL EXPERIENCE

Carve Construction Oct. 2021 - Aug. 2022
General Contractor *Boulder, CO*

- Assisted residential remodeling, including cabinetry, trim, drywall, tile, electrical, framing, and finishing.

Keewaydin Dunmore Camp for Boys Jun. 2023 - Aug. 2023
Counselor *Salisbury, VT*

- Mentor for teenage boys in a rustic outdoor setting teaching outdoor skills and fostering a supportive environment.
- Planned and led multi-week, fully off-grid whitewater canoeing expeditions in Maine and Quebec.

AWARDS & HONORS

Mortimore F. Sayre Prize | *Awarded* | Union College May 2026
Awarded to the senior with the best potential for furthering the ideals of the mechanical engineering profession.

General Electric Edison Leadership Prize | *Awarded* | Union College May 2026
Awarded to a senior majoring in mechanical engineering who has demonstrated exceptional leadership qualities.

Tau Beta Pi Honor Society | *Member* | New York Mu Chapter, Union College Nov. 2025

John S. Hadala (1928) Endowed Book Prize | *Awarded* | Union College May 2025
Awarded to the rising senior majoring in mechanical engineering on the basis of academic, personal, and social achievement.

Pi Tau Sigma Honor Society | *Member* | Alpha Alpha Chapter, Union College Mar. 2025

General Electric Edison Book Prize | *Nominated* | Union College May 2024
Awarded to a sophomore majoring in mechanical engineering who has demonstrated exceptional leadership qualities.

TECHNICAL SKILLS

Experimental Methods: planar particle image velocimetry, volumetric particle tracking velocimetry, flow visualization, wind/water tunnel experimentation

Programming & Analysis: MATLAB, Python

CAD: SolidWorks (Certified Professional), Autodesk Inventor (Certified Associate)

Manufacturing: machining, woodworking, power tools

EXTRACURRICULARS

Interests: biking, ultimate frisbee, hiking, skiing, camping, solving Rubik's cubes